

Flexible perovskite/silicon tandem solar cells with 33.6% efficiency

<https://doi.org/10.1038/s41586-025-09849-4>

Received: 11 April 2025

Accepted: 3 November 2025

Published online: 10 November 2025



Shibo Wang^{1,11}, Wenhao Li^{1,11}, Cao Yu^{2,3,11}, Wei Shi^{1,11}, Qian Kang^{4,11}, Fengxian Cao¹, Kun Gao¹, Liu Yang¹, Bowen Yang¹, Jian Zhou³, Shaofei Yang³, Qi Wang³, Qin Fei³, Xi Chen³, Gaoyuan Chen⁵, Pengxu Chen⁶, Zijia Li⁷, Wei-Chih Hsu⁷, Zhongliang Yan⁸, Yang Bai⁸, Wenzhu Liu⁶, Stefaan De Wolf⁹✉, Xinbo Yang^{1,10}✉ & Xiaohong Zhang^{2,10}✉

Flexible solar cells have a transformative potential for niche applications, but they face fundamental challenges in simultaneously achieving high-power conversion efficiency (PCE), extreme mechanical resilience and operational stability^{1–4}. Here we demonstrate a certified 33.6%-efficient flexible perovskite/crystalline silicon (c-Si) tandem solar cell with a record open-circuit voltage (V_{oc}) of 2.015 V, rivalling its rigid counterpart. The flexible tandem retains 91% of its initial PCE after 5,000 cycles under a bending radius (R_b) of 17.6 mm and demonstrates exceptional operational and damp-heat stability, featuring a T_{80} lifetime exceeding 2,000 h under continuous illumination and retaining 90% of its initial PCE after a 1,000 h damp-heat test. This advancement is enabled by the implementation of the reactive-plasma-deposited (RPD) cerium and hydrogen co-doped indium oxide (ICO:H) recombination layer that promotes self-assembled monolayers (SAMs) coverage and interfacial charge transfer, and in situ annealed zinc-doped indium oxide (IZO) front transparent electrode with markedly enhanced optoelectronic and mechanical properties.

Perovskite-based multi-junction solar cells, such as all-perovskite⁵, perovskite/CIGS⁶, perovskite/organic⁷ and perovskite/c-Si tandems⁸, have been extensively explored. Among these, monolithic perovskite/c-Si tandem solar cells, which stack wide-bandgap perovskite top cells on c-Si bottom cells, stand out as a transformative strategy, combining ultrahigh PCE with cost-effective manufacturing^{8–15}. The certified PCEs of rigid perovskite/c-Si tandem have risen from 13.7% in 2013 to 34.9% in 2025 (refs. 16,17), surpassing the Shockley-Queisser limit of single-junction solar cells. Yet flexible tandem counterparts have so far rarely been investigated. A previous study reported one of the first flexible perovskite/c-Si tandems with a low PCE of 22.8% (ref. 18), and recent studies improved it to 26.5% (ref. 19) and 29.88% (ref. 20). Nevertheless, these values significantly lag behind those of their rigid counterparts. In this work, we demonstrate flexible perovskite/c-Si tandems with a certified PCE of 33.6% and a record open-circuit voltage (V_{oc}) of 2.015 V. This breakthrough bridges the performance gap between flexible and rigid tandems, unlocking new frontiers in lightweight, high-efficiency flexible photovoltaics.

Performance of flexible tandems

Figure 1a shows the architecture of our flexible perovskite/c-Si tandem solar cells. The ultrathin and bendable silicon heterojunction (SHJ) bottom cell (about 65 μm) features sub-micrometre textures on both sides,

over which a thick perovskite absorber is coated. Cross-sectional scanning electron microscopy (SEM) imaging shows vertically aligned, compact perovskite grains conformally grown on the submicron-textured SHJ bottom cell (Supplementary Fig. 1). The wide-bandgap perovskite top cell (1.68 eV) is integrated with RPD ICO:H recombination layer and an in situ annealed IZO front electrode. High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) and energy-dispersive X-ray spectroscopy (EDX) confirm the compact perovskite morphology and identify thin functional layers under and on perovskite (for example, ICO:H, C_{60} , SnO_2 , IZO, Ag and MgF_x) (Fig. 1b). A conformal ICO:H layer (around 10 nm) uniformly coats the textured c-Si surface, terminated with an intrinsic amorphous silicon (a-Si:H(i))/phosphorus-doped nanocrystalline silicon oxide (nc-SiO_x:H(n)) stack (Fig. 1c). RPD ICO:H offers high mobility ($> 130 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), low damage and excellent contact quality^{21–23} and have been widely used for high efficiency SHJ solar cells^{24,25}.

The champion flexible tandem achieves a certified PCE of 33.6% with a record V_{oc} of 2.015 V, a high fill factor (FF) of 81.9% and a moderate short-circuit current density (J_{sc}) of 20.36 mA cm^{-2} (Fig. 1d and Supplementary Fig. 2). Stabilized power output under maximum power point (MPP) tracking confirms a high PCE of 33.2% (Fig. 1e). By contrast, the control tandem with conventional ITO recombination layer (20 nm) and non-annealed IZO front electrode exhibits a much lower PCE of 31.52%. The average PCE of 30 flexible tandems reaches 33.2%, in

¹College of Energy, Soochow University, Suzhou, China. ²Institute of Functional Nano & Soft Materials (FUNSOM), Jiangsu Key Laboratory of Advanced Negative Carbon Technologies, Soochow University, Suzhou, China. ³Suzhou Maxwell Technologies, Suzhou, China. ⁴School of Information Science and Technology, Beijing University of Technology, Beijing, China. ⁵Key Laboratory of Intelligent Optoelectronic Devices and Chips of Jiangsu Higher Education Institutions, School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou, China. ⁶Research Center for New Energy Technology, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai, China. ⁷Chint New Energy Technology, Haining, China. ⁸Institute of Technology for Carbon Neutrality, Shenzhen Institute of Advanced Technology (SIAT), Chinese Academy of Sciences, Shenzhen, China. ⁹Center for Renewable Energy and Storage Technologies (CREST), King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia. ¹⁰Suzhou Laboratory, Suzhou, China. ¹¹These authors contributed equally: Shibo Wang, Wenhao Li, Cao Yu, Wei Shi, Qian Kang. ✉e-mail: stefaan.dewolf@kaust.edu.sa; xbyang@suda.edu.cn; xiaohong_zhang@suda.edu.cn

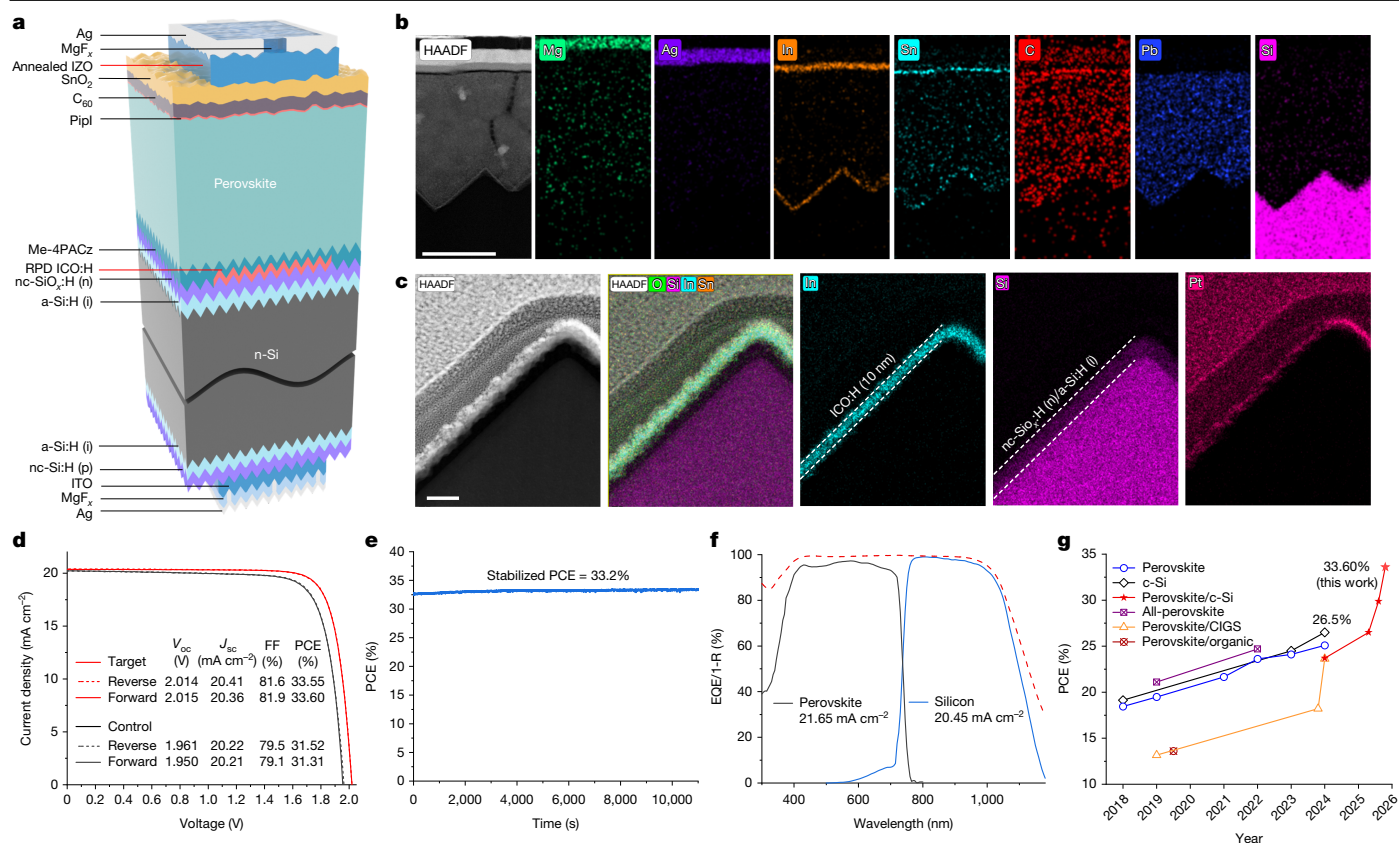


Fig. 1 | Performance of flexible perovskite/c-Si tandems. **a**, Sketch of the flexible tandem with ICO:H recombination layer and in situ annealed IZO front electrode. **b, c**, HAADF-STEM images with EDX mapping of wide-bandgap perovskite top cell (**b**) and ICO:H recombination layer on submicron-textured SHJ bottom cell (**c**). **d**, $J-V$ curve of the champion flexible perovskite/c-Si tandem.

which the ICO:H recombination layer and in situ annealed IZO electrode contributes about 1.5% and about 0.7% absolute PCE gain, respectively (Extended Data Fig. 1). To the best of our knowledge, the V_{oc} of 2.015 V is the highest value reported on monolithic perovskite/c-Si tandem solar cells with an optimal perovskite bandgap of 1.67–1.68 eV. The record V_{oc} arises from dense SAM anchoring on ICO:H, enhancing interfacial charge transfer, and from the ultrathin c-Si bottom cell, which suppresses bulk recombination and deposition damage. The V_{oc} increases by 12 mV when using 65 μm wafers, compared to the industrial 120 μm thickness (Supplementary Fig. 3). Despite the wafer thinning-induced J_{sc} loss in SHJ bottom cell²⁶, the flexible tandem retains a moderate J_{sc} of 20.36 $\text{mA}\cdot\text{cm}^{-2}$, due to high transparency of the in situ annealed IZO electrode and an effective rear reflector^{27,28} (Extended Data Fig. 2). Thus, the rear reflection is maximized and the parasitic absorption loss is minimized, resulting in 21.65 and 20.45 $\text{mA}\cdot\text{cm}^{-2}$ current outputs from the perovskite top cell and SHJ bottom cell (Fig. 1f), respectively. The improved FF can be ascribed to the enhanced electric field at the interconnection junction of ICO:H/SAM, and the reduced sheet resistance (R_s) of in situ annealed IZO front electrode. Above results represent remarkable progress in PCE enhancement for the flexible solar cells (Fig. 1g)^{18,29}, delivering 7.1% absolute PCE improvement compared to the best flexible c-Si solar cells (26.5%). The champion flexible tandem achieves a power-to-weight ratio of 2.6 W/g, among the highest reported for flexible tandem photovoltaics (Supplementary Fig. 4). Its rigid counterpart, employing identical ICO:H recombination layer and in situ annealed IZO front electrode, exhibits marginally lower PCE of 33.15% (Extended Data Fig. 3). Although a higher J_{sc} of 20.94 $\text{mA}\cdot\text{cm}^{-2}$ is obtained with rigid SHJ bottom cell (about 120 μm) because of enhanced light absorption in the near-infrared range, the PCE is compromised

with an aperture area of 1.0 cm^2 , in which the control device is with a conventional ITO recombination layer (20 nm) and non-annealed IZO. **e, f**, Stabilized power output (**e**) and EQE of the champion flexible tandem (**f**). **g**, PCE evolution of various flexible solar cells in recent 10 years. Scale bars, 500 nm (**b**); 50 nm (**c**).

by a lower V_{oc} (1.974 V) and FF (80.2%). We attribute PCE enhancement in flexible tandems to acid-thinning-induced uniform sub- μm textures on the c-Si substrate (Supplementary Fig. 5), which promote conformal SAM anchoring and enhance interfacial charge transfer.

We demonstrate that the tandem solar cells are bendable and cyclically durable down to an R_b of 10 mm over centimetre-scale areas. After 5,000 bending cycles at R_b of 17.6 mm, 15 mm, 12.5 mm and 10 mm, the tandems retain 91.2%, 89.8%, 88.3% and 84.8% of their initial PCE, respectively, demonstrating exceptional mechanical robustness (Fig. 2b). Performance loss correlated primarily with reduced FF (Supplementary Table 1), probably attributable to increased series resistance due to interfacial or IZO front electrode degradation. Control devices exhibit significantly compromised mechanical reliability, retaining only 78.6%, 77.3%, 73.2% and 61.9% of initial PCE under identical R_b . Further reduction of R_b below 10 mm induced mechanical failure within hundreds of cycles, although the minimum R_b for the flexible tandem can reach about 7 mm (Supplementary Fig. 6a). Under compressive stress ($R_b = 17.6$ mm), the tandem maintains 93% of its initial PCE after 5,000 cycles (Supplementary Fig. 6b). The high robustness arises from dense SAM anchoring on ICO:H, which strengthens interfacial bonding, and the superior fracture resistance of the in situ annealed IZO electrode. Nevertheless, the bending tolerance of the flexible tandem remains below the state-of-the-art miniature single-junction perovskite solar cells³⁰ (Supplementary Table 2). Three-point bending test results in Fig. 2c indicate that the presence of the perovskite top cell increases the vertical displacement of the SHJ cell at the cracking moment from 2.04 mm to 2.34 mm, demonstrating enhanced mechanical robustness. This can be ascribed to the shift of the mechanical neutral plane towards the tandem surface, which minimizes the strain on the brittle c-Si substrate¹⁸. Post-bending

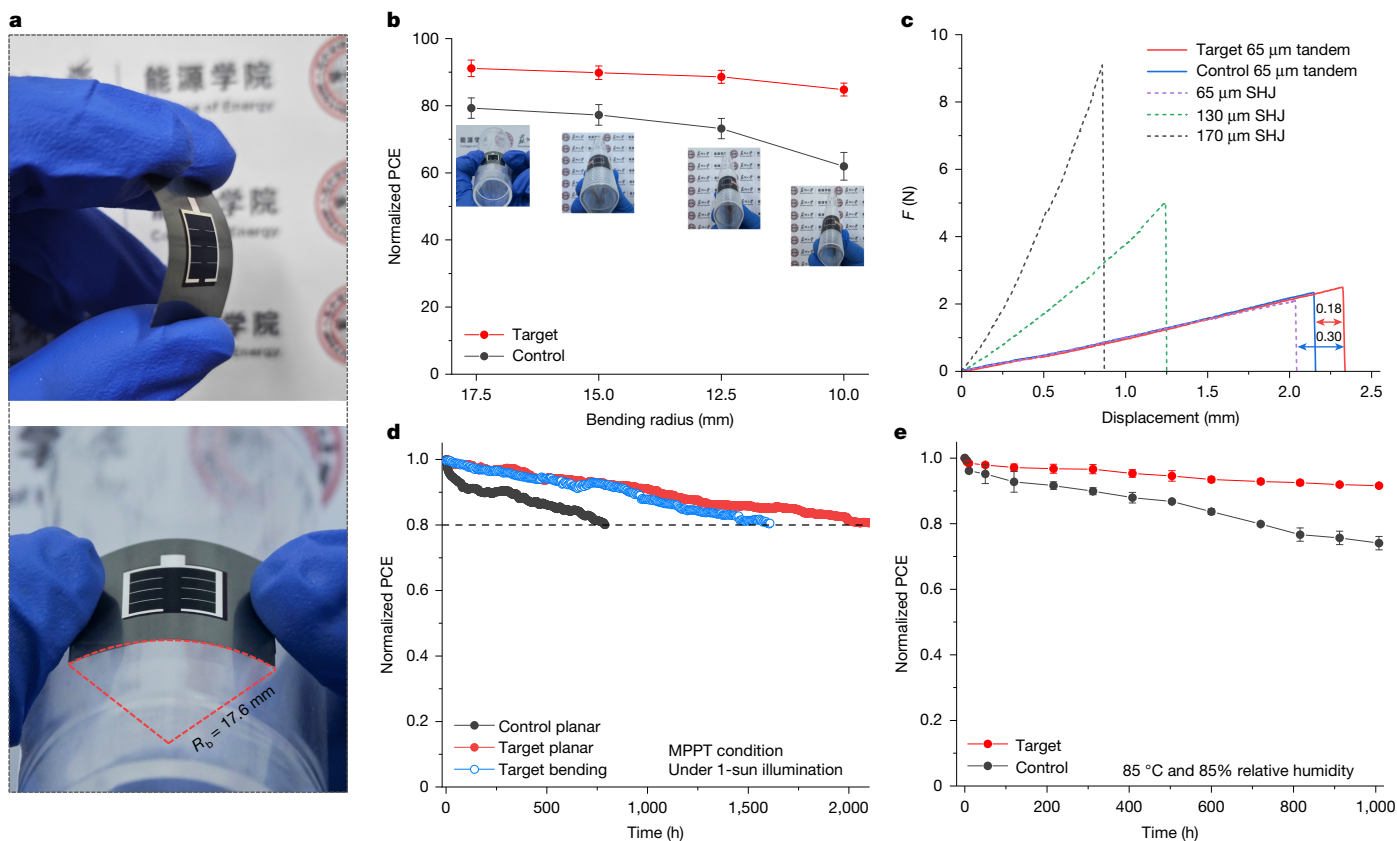


Fig. 2 | Flexibility and stability of flexible tandems. **a**, Photographs of banded tandems with an active area of $10 \times 10 \text{ mm}^2$ on a $25 \times 25 \text{ mm}^2$ c-Si substrate. The bending radius is 17.6 mm. **b**, Bending tests of flexible tandems under tensile stress with different bending radii. The symbols represent the average values from three samples. The absolute starting PCE values are listed in Supplementary Table 1. **c**, Load-vertical displacement (F - D) curves of 65- μm tandems and different thicknesses of SHJ solar cells. **d**, MPP tracking of the

encapsulated flexible tandems under continuous 1-sun illumination in N_2 atmosphere. The absolute starting PCE values are 33.32%, 33.21% and 31.26% for the target planar, target bending and control planar, respectively. **e**, PCE evolution of encapsulated flexible tandems subjected to damp-heat environment. The symbols represent the average values from three samples. The average absolute starting PCE values are 33.14% and 31.03% for the target and control samples, respectively.

characterization shows negligible degradation in standalone SHJ bottom cells (Supplementary Fig. 7), indicating that the PCE loss in flexible tandems originates primarily from the perovskite top cell.

To assess long-term operational stability, encapsulated flexible tandems were subjected to MPP tracking under AM1.5G illumination in a nitrogen ambient atmosphere. The flexible tandems under planar configuration exhibit a high T_{80} lifetime of more than 2,000 h, compared with about 800 h for the control device (Fig. 2d). Under bending stress ($R_b = 17.6 \text{ mm}$), the T_{80} lifetime decreases to about 1,600 h, probably because of strain-induced interfacial failure. The rigid tandem with a thick SHJ bottom cell exhibits a comparable T_{80} lifetime of about 1,900 h (Supplementary Fig. 8). In damp-heat testing (85 °C and 85% relative humidity), the flexible tandem retains 90% of its initial PCE after 1,000 h, outperforming the control device (74%) (Fig. 2e). The operational and damp-heat stability of the flexible tandems is comparable to the best results for their rigid counterparts^{31–33}. The improved stability arises from dense SAM coverage on ICO:H with strong interfacial adhesion, the highly crystalline in situ annealed IZO electrode with better crystallinity and optimized current matching that mitigates charge accumulation in the perovskite top cell³⁴.

ICO:H recombination layer

The recombination layer in perovskite/c-Si tandems plays a pivotal part in bridging the subcells, enabling efficient recombination of photogenerated holes (from perovskite) and electrons (from c-Si) through SAM-modified interfaces^{35,36}. Although conventional sputtered

transparent conductive oxides (TCOs, for example, ITO, IZO) have often been used as recombination layers, their limited SAM coverage density and uniformity, due to sparse surface hydroxyl ($-\text{OH}$) groups, compromise interfacial charge transfer, device performance and operational stability¹⁴. To address this, we implement RPD ICO:H as recombination layer for flexible tandems.

Systematic optimization identified an ICO:H thickness of about 10 nm as optimal for flexible tandems, whereas thicker layers reduced PCE due to lower FF and V_{oc} (Extended Data Fig. 4). Density functional theory (DFT) calculations show that [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid (Me-4PACz) binds more strongly to ICO:H than to ITO, with adsorption energies (E_{ad}) of -3.22 eV and -2.54 eV , respectively (Fig. 3a). We also observe a stronger charge transfer between ICO:H and Me-4PACz. The charge transfer between TCOs and Me-4PACz occurs by forming $\text{In}-\text{O}$ bonds, indicating a strong interaction between ICO:H and Me-4PACz, which forms more stable and stronger chemical bonds when adsorbed on ICO:H.

To compare the adsorption capacity of Me-4PACz on ICO:H and ITO, X-ray photoelectron spectroscopy (XPS) analysis was performed to evaluate their surface stoichiometry before and after Me-4PACz anchoring (Fig. 3b–e). SAMs typically bind to surface $-\text{OH}$ groups on annealing, and RPD ICO:H exhibits a higher $-\text{OH}$ fraction (45.7%) than ITO (30.0%), providing more anchoring sites and stronger interfacial adhesion. The high $-\text{OH}$ groups in ICO:H originate from water vapour used for hydrogen doping during deposition. By investigating the XPS core-level spectra of P 2p and N 1s of ICO:H and ITO with Me-4PACz (Supplementary Fig. 9), we confirm the successful anchoring of Me-4PACz

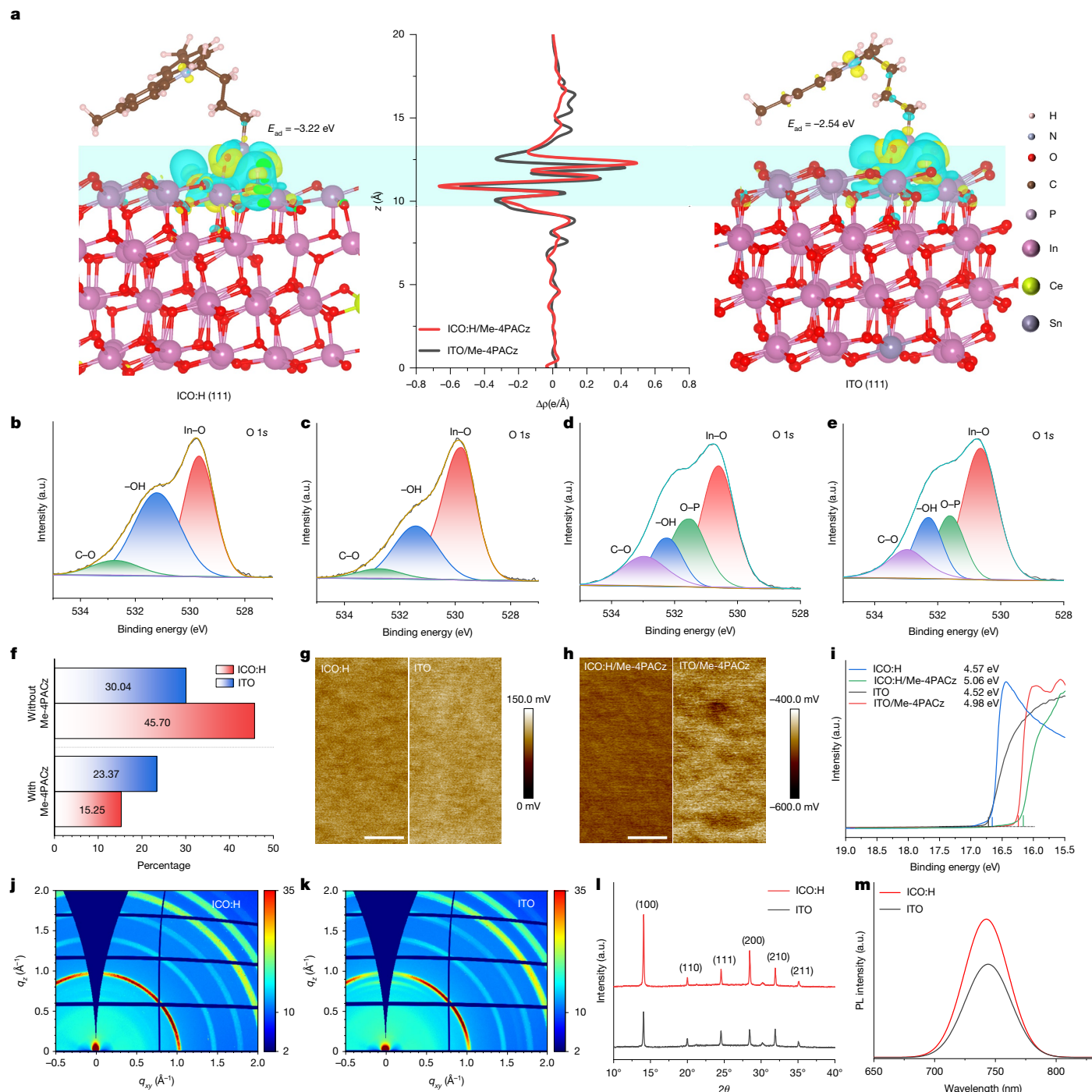


Fig. 3 | Adsorption of Me-4PACz on ICO:H. **a**, DFT calculation of Me-4PACz adsorbed on ICO:H and ITO, and charge transfer between Me-4PACz and ICO:H (ITO). **b–e**, XPS core-level spectra of O 1s for ICO:H and ITO before and after Me-4PACz anchoring. **f**, Calculated proportion of -OH group for ICO:H and ITO before and after Me-4PACz anchoring. **g–i**, KPFM images (**g,h**) and UPS

cut-offs (**i**) of ICO:H and ITO with and without Me-4PACz anchoring. **j,k**, GIWAX images of the buried surfaces of perovskite grown on ICO:H and ITO. **l,m**, XRD patterns (**l**) and photoluminescence spectra (**m**) of perovskites grown on ICO:H and ITO. a.u., arbitrary units. Scale bar, 1 μm (**g,h**).

on them. After Me-4PACz anchoring, the -OH proportion decreases to 15.3% for ICO:H and 23.4% for ITO (Fig. 3f), confirming stronger bonding on ICO:H. The higher P-O/In-O ratio (0.56 compared with 0.43) further indicates higher Me-4PACz coverage density on ICO:H. Atomic force microscopy images show negligible change in surface roughness (Supplementary Fig. 10), suggesting uniform monolayer formation without residue.

Kelvin probe force microscopy (KPFM) was further conducted to analyse the surface potential uniformity of ICO:H and ITO before and

after Me-4PACz anchoring (Fig. 3g,h). The average potential differences of ICO:H before and after Me-4PACz anchoring are consistently lower than that of ITO (Supplementary Fig. 11), indicating a more uniform surface potential distribution of ICO:H. ICO:H also shows a lower surface potential both before and after anchoring (Supplementary Fig. 12), implying a higher work function. Ultraviolet photoelectron spectroscopy (UPS) confirms this, with ICO:H exhibiting a work function of 5.06 eV versus 4.98 eV for ITO after Me-4PACz anchoring (Fig. 3i). The higher work function facilitates the transfer of charge carriers at

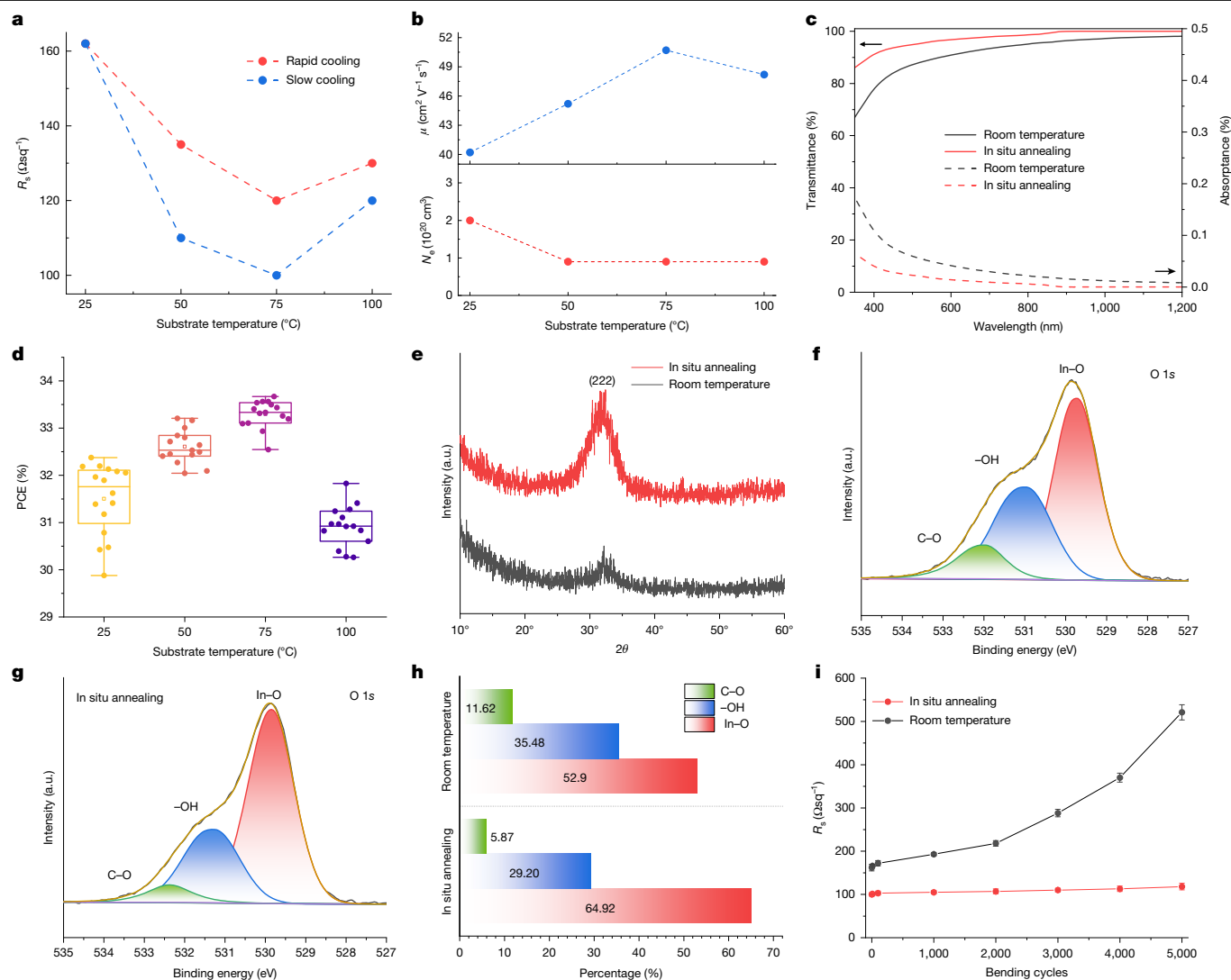


Fig. 4 | In situ annealed IZO front electrode. **a, b**, Effect of in situ annealing temperatures on the R_s , μ_e and N_c of IZO. **c**, The UV-Vis-NIR spectra of IZO with and without in situ annealing, in which the solid and dashed lines represent the transmittance and absorbance curves, respectively. **d**, Effect of in situ annealing temperatures on the PCE of flexible tandems. Fifteen devices were fabricated for each condition. The top lines, bottom lines, lines in the box, square and boxes

represent maximum values, minimum values, median values, mean values and 25–75% distributions, respectively. **e**, XRD patterns of IZO with and without in situ annealing; **f, g**, XPS core-level spectra of the O 1s in IZO without and with in situ annealing. **h**, Proportion of In–O, C–O and O^{2-} . **i**, R_s of IZO films after undergoing 5,000 cycles of bending tests ($R_b = 17.6$ mm). The symbols represent the average values from three samples. a.u., arbitrary units.

the perovskite/Me-4PACz/ICO:H interface, enhancing recombination efficiency and tandem performance.

To examine how Me-4PACz coverage density on TCOs affects perovskite crystallinity, we further analysed the buried interface. The SEM images indicate that perovskites grown on ICO:H show fewer voids and less residual PbI_2 than those on ITO, with large and vertically oriented grains (Supplementary Fig. 13). Grazing-incidence wide-angle X-ray scattering (GIWAXS) shows a stronger (100) diffraction peak for perovskites grown on ICO:H, whereas ITO shows an additional PbI_2 ring at $q = 0.9 \text{ \AA}^{-1}$ (Fig. 3j, k). X-ray diffraction (XRD) confirms enhanced (100) peak intensity for ICO:H-grown perovskites (Fig. 3l), consistent with improved crystallinity. Correspondingly, photoluminescence spectra exhibit higher emission intensity (Fig. 3m), indicating a better perovskite quality. Single-junction wide-bandgap perovskite top cell on textured c-Si substrate with a thick ICO:H bottom electrode (150 nm) achieves a remarkable PCE of 22.85% with a high V_{oc} of 1.27 V (Supplementary Fig. 14), confirming the high-quality perovskite grown on the ICO:H. These results indicate that the dense Me-4PACz layer on ICO:H can promote superior perovskite growth,

underpinning the enhanced performance and flexibility of the tandem devices.

In situ annealed IZO front electrode

The front transparent electrode is an important component for tandems, requiring high lateral conductivity, broadband transparency and minimal deposition damage to underlying layers. However, conventional TCOs are brittle and prone to cracking under tensile stress, compromising the flexibility and reliability of the tandems. Sputtered IZO deposited at room temperature is widely used for perovskite/c-Si tandems owing to its excellent optoelectronic properties without high-temperature annealing, as prohibited by the temperature-sensitive perovskite top cell. Here, we introduce an in situ annealing process during IZO deposition, which simultaneously enhances its electrical conductivity, optical transparency and mechanical robustness for flexible tandems.

To evaluate the effect of in situ annealing on IZO properties, we varied the substrate temperature from 25 °C to 100 °C. As shown in Fig. 4a,

the R_s of IZO (40 nm) decreases from $160\ \Omega\text{sq}^{-1}$ to $100\ \Omega\text{sq}^{-1}$ at an optimal $75\ ^\circ\text{C}$, with slow post-deposition cooling further improving conductivity. Hall measurements confirm that carrier mobility (μ_e) increases from $40\ \text{cm}^2\text{V}^{-1}\text{s}^{-1}$ to $50.7\ \text{cm}^2\text{V}^{-1}\text{s}^{-1}$, whereas carrier concentration (N_e) decreases from $2 \times 10^{20}\ \text{cm}^{-3}$ to $1 \times 10^{20}\ \text{cm}^{-3}$ (Fig. 4b). Consequently, in situ annealed IZO exhibits higher conductivity and lower optical absorption (Fig. 4c), which contribute to the enhanced performance of the flexible tandems. The average PCE of the flexible tandems increases from 31.7% to 33.4% (Fig. 4d), which can be mainly attributed to the improved J_{sc} and FF (Extended Data Fig. 5). XRD shows stronger (222) diffraction peaks with in situ annealing at $75\ ^\circ\text{C}$ (Fig. 4e), consistent with TEM with selected area electron diffraction and atomic force microscopy observations of larger crystallites embedded in an amorphous matrix and reduced grain-boundary scattering (Supplementary Figs. 15 and 16). XPS analysis shows oxygen vacancies decreasing from 35.5% to 29.2% with in situ annealing (Fig. 4f–h). The growth of grains during in situ annealing results in the filling of oxygen vacancies, probably because of the flow of interstitial oxygen and the increased oxygen atoms in the air entering the film³⁷. The reduced N_e contributes to the higher transparency and lower absorbance of IZO, and hence a higher J_{sc} on the flexible tandems.

The mechanical robustness of in situ annealed IZO was evaluated by cyclic bending (Fig. 4i). After 5,000 cycles, the average R_s increased only slightly from $99\ \Omega\text{sq}^{-1}$ to $118\ \Omega\text{sq}^{-1}$, whereas non-annealed IZO degraded severely from $162\ \Omega\text{sq}^{-1}$ to $521\ \Omega\text{sq}^{-1}$. We observe cracks on the non-annealed IZO, whereas no cracks on the in situ annealed IZO (Supplementary Fig. 17). The enhanced mechanical property of in situ annealed IZO can be related to the higher crystallinity, which is proven to be the primary factor in determining the mechanical strength of ITO³⁸. The higher crystallinity and the better crystalline quality, the higher mechanical reliability. Moreover, in situ annealed IZO improves tandem stability by mitigating halogen diffusion. As we know the halogens (for example, I^- and Br^-) can diffuse through the grain boundaries and defects in TCOs and react with the silver electrode, resulting in the irreversible degradation of the perovskite solar cells^{39–42}. Enhanced crystallinity and fewer oxygen vacancies reduce pathways for halide migration, thereby preventing Ag electrode corrosion during damp-heat testing. Overall, in situ annealed IZO simultaneously improves conductivity, mechanical reliability and long-term stability of flexible tandem solar cells.

This work represents remarkable advance in flexible photovoltaic technology, demonstrating that perovskite/c-Si tandems can simultaneously achieve high PCE, mechanical durability and long-term stability. By integrating interfacial and electrode engineering at the material level, it bridges the gap between high-performance rigid tandems and flexible, deployable solar systems, opening new opportunities for lightweight energy solutions in aerospace, wearable, mobile energy, building-integrated photovoltaics and Internet of Things applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09849-4>.

- Huang, S. et al. A review on flexible solar cells. *Sci. China Mater.* **67**, 2717–2736 (2024).
- Li, S., Li, Z., Wan, X. & Chen, Y. Recent progress in flexible organic solar cells. *eScience* **3**, 100085 (2023).
- Tian, R., Zhou, S., Meng, Y., Liu, C. & Ge, Z. Material and device design of flexible perovskite solar cells for next-generation power supplies. *Adv. Mater.* **36**, 2311473 (2024).
- Lee, D. S. et al. Overcoming stability limitations of efficient, flexible perovskite solar modules. *Joule* **8**, 1380–1393 (2024).
- Liu, Z. et al. All-perovskite tandem solar cells achieving >29% efficiency with improved (100) orientation in wide-bandgap perovskites. *Nat. Mater.* **24**, 252–259 (2025).
- Pei, F. et al. Inhibiting defect passivation failure in perovskite for perovskite/Cu(In,Ga)Se₂ monolithic tandem solar cells with certified efficiency 27.35%. *Nat. Energy* **10**, 824–835 (2025).

- Brinkmann, K. O. et al. Perovskite–organic tandem solar cells. *Nat. Rev. Mater.* **9**, 202–217 (2024).
- Ugur, E. et al. Enhanced cation interaction in perovskites for efficient tandem solar cells with silicon. *Science* **385**, 533–538 (2024).
- Chen, Y. et al. Nuclei engineering for even halide distribution in stable perovskite/silicon tandem solar cells. *Science* **385**, 554–560 (2024).
- Mariotti, S. et al. Interface engineering for high-performance, triple-halide perovskite–silicon tandem solar cells. *Science* **381**, 63–69 (2023).
- Chin, X. Y. et al. Interface passivation for 31.25%-efficient perovskite/silicon tandem solar cells. *Science* **381**, 59–63 (2023).
- Liu, J. et al. Perovskite/silicon tandem solar cells with bilayer interface passivation. *Nature* **635**, 596–603 (2024).
- Kim, D. et al. Efficient, stable silicon tandem cells enabled by anion-engineered wide-bandgap perovskites. *Science* **368**, 155–160 (2020).
- Aydin, E. et al. Enhanced optoelectronic coupling for perovskite/silicon tandem solar cells. *Nature* **623**, 732–738 (2023).
- Liu, J. et al. Efficient and stable perovskite–silicon tandem solar cells through contact displacement by MgF_2 . *Science* **377**, 302–306 (2022).
- NREL. *Best Research-Cell Efficiency Chart*. Photovoltaic Research. <https://www.nrel.gov/pv/cell-efficiency.html>.
- Mailoa, J. P. et al. A 2-terminal perovskite/silicon multijunction solar cell enabled by a silicon tunnel junction. *Appl. Phys. Lett.* **106**, 121105 (2015).
- Wang, X. et al. Ultrathin ($\sim 30\ \mu\text{m}$) flexible monolithic perovskite/silicon tandem solar cell. *Sci. Bull.* **69**, 1887–1894 (2024).
- Shishido, H. et al. High-efficiency perovskite/silicon tandem solar cells with flexibility. *Sol. RRL* **9**, 2400899 (2025).
- Sun, Y. et al. Flexible perovskite/silicon monolithic tandem solar cells approaching 30% efficiency. *Nat. Commun.* **16**, 5733 (2025).
- Shi, J., Meng, F., Bao, J., Liu, Y. & Liu, Z. Surface scattering effect on the electrical mobility of ultrathin Ce doped In_2O_3 film prepared at low temperature. *Mater. Lett.* **225**, 54–56 (2018).
- Yu, C. et al. 26.6%-Efficiency Silicon heterojunction solar cell with high-quality cerium and hydrogen codoped indium oxide transparent electrode. *ACS Energy Lett.* **10**, 2503–2511 (2025).
- Kobayashi, E., Watabe, Y. & Yamamoto, T. High-mobility transparent conductive thin films of cerium-doped hydrogenated indium oxide. *Appl. Phys. Express* **8**, 015505 (2015).
- Ru, X. et al. Silicon heterojunction solar cells achieving 26.6% efficiency on commercial-size p-type silicon wafer. *Joule* **8**, 1092–1104 (2024).
- Lin, H. et al. Silicon heterojunction solar cells with up to 26.81% efficiency achieved by electrically optimized nanocrystalline-silicon hole contact layers. *Nat. Energy* **8**, 789–799 (2023).
- Li, Y. et al. Flexible silicon solar cells with high power-to-weight ratios. *Nature* **626**, 105–110 (2024).
- Tockhorn, P. et al. Nano-optical designs for high-efficiency monolithic perovskite–silicon tandem solar cells. *Nat. Nanotechnol.* **17**, 1214–1221 (2022).
- Bush, K. A. et al. 23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability. *Nat. Energy* **2**, 17009 (2017).
- Hu, S. et al. Flexible perovskite-based multiple-junction photovoltaics. *Joule* **9**, 101870 (2025).
- Li, Z. et al. Hyperbranched polymer functionalized flexible perovskite solar cells with mechanical robustness and reduced lead leakage. *Nat. Commun.* **14**, 6451 (2023).
- Kan, C. et al. Efficient and stable perovskite–silicon tandem solar cells with copper thiocyanate-embedded perovskite on textured silicon. *Nat. Photon.* **19**, 63–70 (2025).
- Aydin, E. et al. Pathways toward commercial perovskite/silicon tandem photovoltaics. *Science* **383**, eadh3849 (2024).
- Duan, L. et al. Stability challenges for the commercialization of perovskite–silicon tandem solar cells. *Nat. Rev. Mater.* **8**, 261–281 (2023).
- Jošt, M. et al. 21.6%-Efficient monolithic perovskite/Cu(In,Ga)Se₂ tandem solar cells with thin conformal hole transport layers for integration on rough bottom cell surfaces. *ACS Energy Lett.* **4**, 583–590 (2019).
- De Bastiani, M. et al. Recombination junctions for efficient monolithic perovskite-based tandem solar cells: physical principles, properties, processing and prospects. *Mater. Horiz.* **7**, 2791–2809 (2020).
- Isikgor, F. H. et al. Molecular engineering of contact interfaces for high-performance perovskite solar cells. *Nat. Rev. Mater.* **8**, 89–108 (2023).
- Liu, H. et al. Effect of annealing treatment on properties of Ce-doped indium oxide (ICO) transparent conductive oxide films. *J. Mater. Sci. Mater. Electron.* **34**, 840 (2023).
- Kim, J.-H. & Park, J.-W. Improving the flexibility of large-area transparent conductive oxide electrodes on polymer substrates for flexible organic light emitting diodes by introducing surface roughness. *Org. Electron.* **14**, 3444–3452 (2013).
- Kato, Y. et al. Silver iodide formation in methyl ammonium lead iodide perovskite solar cells with silver top electrodes. *Adv. Mater. Interfaces* **2**, 1500195 (2015).
- Domanski, K. et al. Not all that glitters is gold: metal-migration-induced degradation in perovskite solar cells. *ACS Nano* **10**, 6306–6314 (2016).
- Wu, S. et al. A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells. *Nat. Commun.* **10**, 1161 (2019).
- Xiao, M. et al. Engineering amorphous–crystallized interface of ZnO, barriers for stable inverted perovskite solar cells. *Adv. Mater.* **35**, 2301684 (2023).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025

Methods

Materials

Anhydrous dimethyl sulfoxide (DMSO), dimethylformamide (DMF) and ethyl acetate were purchased from Sigma Aldrich. Formamidine iodide (FAI, 99.99%) and methylammonium bromide (MABr, 99.99%) were purchased from GreatCell Solar Materials. C_{60} was bought from Nano-C. Piperazinium iodide (PipI), caesium iodide (CsI), Me-4PACz, lead iodide (PbI_2), lead bromide ($PbBr_2$) and 1,6-hexylenediphosphonic acid (6-dPA) were purchased from TCI Chemicals. Tetrakis(dimethylamino)tin(IV) (TDMASn) was purchased from Ai Mou Yuan Scientific Equipment. The 3-inch IZO (In_2O_3 with 10 wt% ZnO) and ITO (In_2O_3 with 10 wt% SnO_2) targets were purchased from ZhongNuo Advanced Material Technology. All materials were used as received unless specified.

Device fabrication

N-type (100)-oriented Czochralski (C_z) wafers (1–5 Ω cm, 130 μ m) were used for the fabrication of bendable and rigid SHJ bottom cells. A wet-chemical process, including saw-damage removal, texturing and cleaning, was first applied to the as-cut wafers. The standard wafers were thinned to 65 μ m using a mixed acid solution (HNO_3/HF). The edge-blunting process involves chemical etching (10 vol% HF :90 vol% HNO_3) of the marginal region (about 2 mm) of the 65- μ m textured wafers for 90 s at room temperature, as detailed in our previous publication⁴³. The concentrations of HF and HNO_3 were 49% and 68%, respectively, diluted in water. The SHJ bottom cells were fabricated as reported in our previous publication⁴⁴. Intrinsic hydrogenated amorphous silicon (a-Si:H, 5 nm) was deposited by plasma-enhanced chemical vapour deposition on both sides of submicro-textured wafers. Subsequently, n-type nanocrystalline silicon oxide (nc- $SiO_x:H(n)$, 24 nm) and p-type nanocrystalline silicon (nc-Si:H(p), 30 nm) were deposited at the front and rear sides, respectively. Then ICO:H or ITO recombination layer was deposited onto the nc- $SiO_x:H(n)$ layer through a shadow mask, defining the active area (1×1 cm²). The ICO:H and ITO were deposited by RPD and magnetron sputtering, respectively. The ICO:H films were deposited under a fixed H_2/Ar ratio (0.63%) and chamber pressure (0.4 Pa) using CeO_2 -doped In_2O_3 tablet (3 wt%). Water vapour with a low flow rate of 15 sccm was used for hydrogen doping. The back side was prepared by stacking sputtered ITO (20 nm) through a shadow mask and then screen-printing Ag fingers with 1 mm spacing. After curing at 200 °C for 15 min in the air, 150 nm MgF_2 layers were thermally evaporated in the active area. Finally, the rear contact was finished with a 500 nm thermally evaporated Ag electrode. The SHJ cells were laser cut down to a 25×25 mm² size with an active area of 1×1 cm² in the centre. The bottom cells were cut in the same orientation and direction, eliminating the anisotropy effect on the mechanical durability. All the equipment for silicon thin-film and TCOs deposition was produced by Suzhou Maxwell Technologies.

The tandems were fabricated on 65- μ m-thick SHJ bottom cells with edge blunting to enhance bend tolerance. The SHJ bottom cells were first annealed at 180 °C for 20 min in a nitrogen atmosphere on a hotplate. Me-4PACz solution in ethanol (1.0 mg ml⁻¹) with 20 vol% 6-dPA (0.75 mg ml⁻¹ in ethanol) was subjected to spin-coating in a glovebox at 3,000 rpm for 30 s and annealed at 100 °C for 10 min. The wide-bandgap perovskite precursor ($CS_{0.05}MA_{0.15}FA_{0.80}Pb(I_{0.75}Br_{0.25})_3$, 1.7 M) was prepared by mixing FAI, CsI, MABr, $PbBr_2$ and PbI_2 in mixed DMF and DMSO solvent with a volume ratio of 4:1 and shaken overnight at 60 °C. The perovskite films were subjected to spin-coating at 2,000 rpm for 25 s followed by 4,500 rpm for 25 s. Ethyl acetate (300 μ l) was dropped on the centre of the substrates 10 s before the end of the spin-coating. The perovskite films were then annealed at 100 °C for 30 min on a hot plate. Then, a 0.3 mg ml⁻¹ solution of PipI in IPA was subjected to spin-coating at 5,000 rpm for 30 s and annealed at 100 °C for 10 min. IPA was subjected to spin-coating to wash off the excess PipI on the

perovskite surface, using the same spin-coating recipe. The spin coater (Brand-REESEEN, PvS-mini9) was produced by Jiangyin J. Wanjia Technology. The samples were then transferred to a thermal evaporation chamber for the deposition of C_{60} (10 nm) at an evaporation rate of 0.2 $\text{\AA s}^{-1}$. SnO_2 buffer layer (15 nm) was then deposited by atomic layer deposition (E200SP, SUPRALD) at 100 °C using TDMASn and deionized water as precursors. Subsequently, 40 nm IZO was deposited by sputtering through a shadow mask. For the in situ annealing process, the substrate was heated to the target temperature (25–100 °C) and stabilized for 10 min, as monitored by temperature measurement strips. Then IZO was sputtered using the same recipe, followed by slow cooling to room temperature in the chamber or rapid cooling in the air. The active area of 1 cm² was defined by a thermally evaporated Ag electrode (500 nm) with a deposition rate of 5 $\text{\AA s}^{-1}$ using a hard mask. Finally, 120 nm MgF_x was evaporated on IZO as an antireflection layer (1 $\text{\AA s}^{-1}$).

For tandem encapsulation, the front and rear silver electrodes were first connected using conductive copper strips. It is essential to ensure a robust electrical contact between the silver electrode and the conductive copper strips, which extend from the sealed section to serve as electrodes for stability testing. Subsequently, the tandem device was sandwiched between two thick glasses (1.1 mm), and the edges were sealed with butyl rubber, followed by curing with ultraviolet light. The glass seal requires a drying period of 1 day. All seal procedures were conducted in a nitrogen atmosphere. All flexible tandems underwent identical encapsulation before bending tests. Poly(ethylene-octene) (POE) films laminated on both sides via vacuum lamination (100 °C, 20 min) to ensure complete cross-linking. A fluoropolymer backsheet (KPF) served as the rear moisture barrier (water vapour transmission rate $< 5 \times 10^{-4}$ g m⁻² day⁻¹). During bent-state MPP tracking, KPF was additionally applied as the top barrier. For mechanical bending durability tests, polyvinylidene fluoride (PVDF) replaced KPF as the top encapsulant to enhance flexibility.

Characterizations

The thickness of thin films was measured by an ellipsometer (SE 800 PV, Sentech). Load-vertical displacement curves of SHJ and tandem devices (2.5×2.5 cm²) were measured using a commercial Discovery DMA 850 instrument. The structural properties of perovskite were measured by XRD (D8 ADVANCE, Bruker) using Cu K α radiation in the conventional 2θ - θ test mode. The optical transmittance of IZO was measured using a spectrophotometer (LAMBDA 1050, PerkinElmer). SEM was conducted using a field-emission scanning electron microscope (JSM-IT800, JEOL) with an accelerator voltage of 5 kV. XPS and UPS spectra were obtained using a Thermo Fisher Scientific XPS spectrometer (ESCALAB 250Xi), which used a beam of X-ray or UV photons to eject electrons from the near-surface region of samples in vacuum. UPS measurements were carried out using a monochromatized HeI (21.2 eV) photon source, with the sample biased by 5 eV to observe the low-energy cutoff. Atomic force microscopy and Kelvin probe force microscopy measurements were performed using a Bruker Dimension Icon atomic force microscope with an intelligent mode at a scan rate of 0.977 Hz. The resistivity, carrier density and mobility of IZO were measured by Hall effect measurement that performed on a Lake Shore analyser. The HAADF-STEM images with EDX mapping were obtained on FEI Tecnai F30 TEM operating at 300 kV. The cerium element (3.0 wt%) is not shown in the EDX image because of the detection limit. Steady-state photoluminescence spectra were measured with a photoluminescence spectrometer system (FLS1000, Edinburgh) with an excitation light of 450 nm and a pulsed excitation laser of 485 nm, respectively. Crystallinity of perovskite was measured by GIWAXS (BL14B1 beamline of the Shanghai Synchrotron Radiation Facility using X-ray with a wavelength of 0.6887 $\text{\AA}$).

Illuminated J - V curve of tandem solar cells was measured by an LED-based solar simulator (Wavelabs Sinus 300) with AM1.5G irradiance spectrum using a Keithley 2400 source meter. The parameters were -0.02 V to 2 V range, with a 0.02 V step, 20 ms integration time and 20 ms setting time. The shadow mask defines the active area for tandem

solar cells. The calibration of the flexible tandem was performed at China Photovoltaic Testing Center, which is internationally recognized by the International Laboratory Accreditation Cooperation, instead of the China National Accreditation Service. External quantum efficiency (EQE) measurements were performed by the QE-R3018 system from EnliTech in the air. During the measurement of top perovskite cells and bottom silicon cells, the tandem device was saturated with near-infrared (NIR) lights (850 nm) and blue lights (455 nm), respectively. The operational stability of the encapsulated tandems was evaluated using stability test system supplied by PURI Materials (PURI 2400-8Q) under AM1.5G illumination from LED lamp in a nitrogen ambient. The bending durability tests under tensile stress were performed for the tandems with an active area of $10 \times 10 \text{ mm}^2$ on $25 \times 25 \text{ mm}^2$ c-Si substrate. The R_b was defined as the distance between the convex top of the tandem and the imaginary centre of the circle it belongs to. The bending cycle measurements for flexible tandems were performed using a custom-made platform in nitrogen glovebox, with adjustable bend radius and speed from Liaoning SHUZE Solar Photovoltaic Technology.

Data availability

All data are available in the main text or the Supplementary Information. Further data are available from the corresponding authors upon reasonable request.

43. Liu, W. et al. Flexible solar cells based on foldable silicon wafers with blunted edges. *Nature* **617**, 717–723 (2023).

44. Yu, C. et al. Industrial-scale deposition of nanocrystalline silicon oxide for 26.4%-efficient silicon heterojunction solar cells with copper electrodes. *Nat. Energy* **8**, 1375–1385 (2023).

Acknowledgements X.Y. acknowledges financial support from the National Natural Science Foundation of China (no. 62174114 and no. U24A2063), the National Key R&D program of China (no. 2022YFB4200203), the Department of Science and Technology of Jiangsu Province (BE2022023). X.Z. acknowledges financial support from the Key Project of Jiangsu Provincial Basic Research Plan (no. BK20243031), Suzhou Laboratory–GCL–Soochow University Joint Laboratory for Perovskite Photovoltaics and the Collaborative Innovation Center of Suzhou Nano Science and Technology.

Author contributions X.Y. conceived the idea, designed the overall experiments and led the project. X.Y. and S.W. wrote the paper. S.W., W. Li, C.Y. and W.S. fabricated the tandem devices and performed the characterization and analysis. Q.K. performed the TEM characterization and analysis. F.C., K.G., L.Y. and B.Y. contributed to characterizations and data interpretation. J.Z., S.Y., Q.W., Q.F. and X.C. designed and fabricated the silicon bottom cells. G.C. performed the molecular dynamics simulations. P.C. and W. Liu helped with the three-point bending test and analysis. Z.L., W.-C.H., Z.Y. and Y.B. helped with the stability tests. S.D.W., X.Y. and X.Z. supervised this project. All authors contributed to the discussion of the results and revision of the paper.

Competing interests J.Z., S.Y., Q.W., Q.F. and X.C. are employees of Suzhou Maxwell Technologies. Z.L. and W.-C.H. are employees of Chint New Energy Technology. The other authors declare no competing interests.

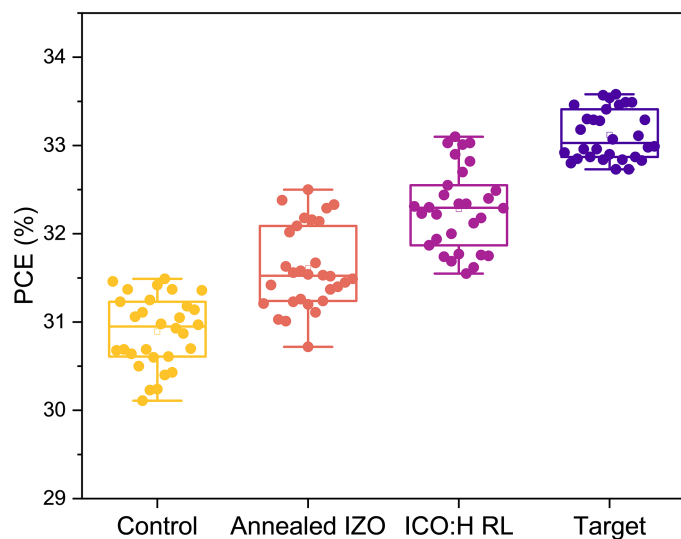
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-025-09849-4>.

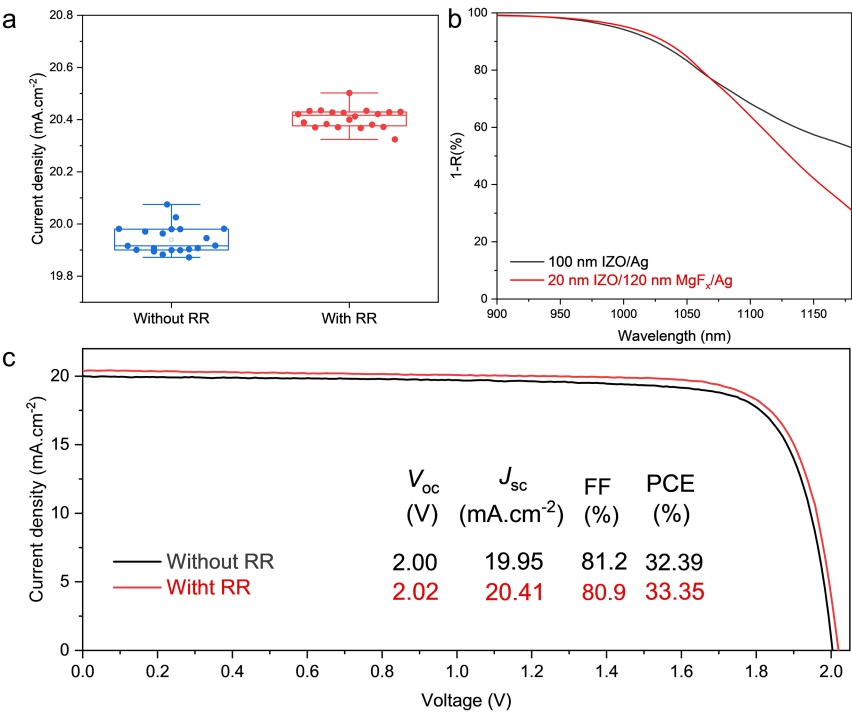
Correspondence and requests for materials should be addressed to Stefaan De Wolf, Xinbo Yang or Xiaohong Zhang.

Peer review information *Nature* thanks the anonymous reviewers for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at <http://www.nature.com/reprints>.

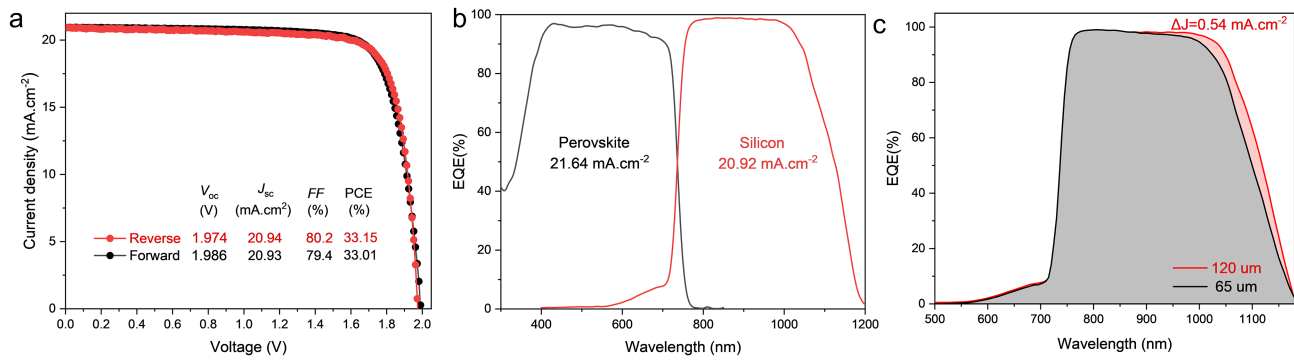


Extended Data Fig. 1 | Flexible tandem performance with recombination layer and front electrode engineering. PCE distribution of flexible tandems with *in-situ* annealed IZO electrode, ICO:H recombination layer, and *in-situ* annealed IZO + ICO:H recombination layer. The control device is with conventional ITO recombination layer (20 nm) and IZO front electrode without *in-situ* annealing. Thirty devices were fabricated for each condition. The top lines, bottom lines, lines in the box, square and boxes represent maximum values, minimum values, median values, mean values and 25–75% distributions, respectively.



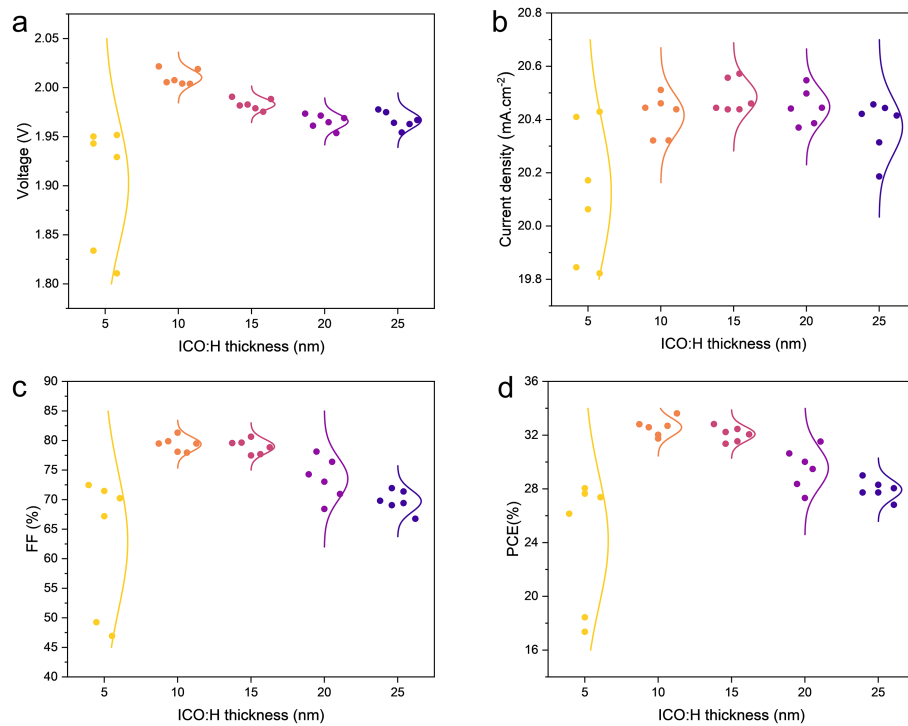
Extended Data Fig. 2 | Flexible tandem performance with and without rear reflector (RR). **a**, Box plot of J_{sc} for devices with and without RR, indicating an absolute J_{sc} gain of 0.4 mA.cm^{-2} . Twenty devices were fabricated for each condition. The top lines, bottom lines, lines in the box, square and boxes

represent maximum values, minimum values, median values, mean values and 25–75% distributions, respectively. **b**, 1-R of the flexible tandem with and without RR within the wavelength range of 900–1180 nm. **c**, J - V curves for the flexible tandems with and without RR.

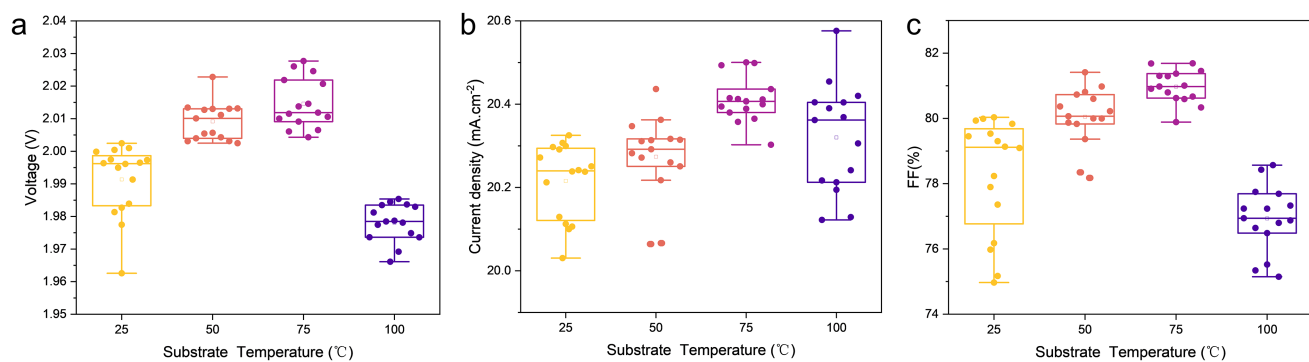


Extended Data Fig. 3 | Performance of rigid tandem. **a**, The J - V curve and **b**, EQE curve of rigid tandem with identical ICO:H recombination layer and *in-situ* annealed IZO on a thick SHJ bottom cell ($\sim 120 \mu\text{m}$). **c**, EQE curves of the SHJ

subcell with different substrate thickness, demonstrating the reduced EQE response in the NIR when decreasing the substrate thickness.



Extended Data Fig. 4 | Effect of ICO:H recombination layer thickness on the performance of flexible tandems. a, V_{oc} . b, FF. c, J_{sc} . d, PCE. The curves are normal distributions from six devices.



Extended Data Fig. 5 | Effect of *in-situ* annealing temperature on the performance of flexible tandems with ICO:H recombination layer. a, V_{oc} . b, J_{sc} . c, FF. Fifteen devices were fabricated for each condition. The top lines,

bottom lines, lines in the box, square and boxes represent maximum values, minimum values, median values, mean values and 25–75% distributions, respectively.